Precambrian Research 164 (2008) 40–49

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Columbia revisited: Paleomagnetic results from the 1790 Ma colider volcanics (SW Amazonian Craton, Brazil)

Franklin Bispo-Santos[a], Manoel S. D'Agrella-Filho[a,*], Igor I.G. Pacca[a], Liliane Janikian[a], Ricardo I.F. Trindade[a], Sten-Ake Elming[b], Jesué A. Silva[c], Márcia A.S. Barros[d], Francisco E.C. Pinho[d]

[a] Instituto de Astronomia, Geofisica e Ciências Atmosféricas, Universidade de São Paulo, Rua do Matão, 1226, Cidade Universitária, 05508-090 São Paulo, Brazil
[b] Department of Applied Geophysics, Luleå University of Technology, 971 87 Luleå, Sweden
[c] Companhia Matogrossense de Mineração, METAMAT, Av. Gonçalo Antunes de Barros, 2970, Planalto, 78050-300 Cuiabá, Mato Grosso, Brazil
[d] Department of Mineral Resources, UFMT, Av. Fernando Correa s/n, 78060-900 Cuiabá, Mato Grosso, Brazil

ARTICLE INFO

Article history:
Received 10 September 2007
Received in revised form 12 February 2008
Accepted 9 March 2008

Keywords:
Paleomagnetism
Paleoproterozoic
Colider Suite
Amazonian Craton
Columbia supercontinent

ABSTRACT

In an attempt to improve our understanding of the Paleoproterozoic geodynamic evolution, a paleomagnetic study was performed on 10 sites of acid volcanic rocks of the Colider Suite, southwestern Amazonian Craton. These rocks have a well-dated zircon U–Pb mean age of 1789 ± 7 Ma. Alternating field and thermal demagnetization revealed northern (southern) directions with moderate to high upward (downward) inclinations. Rock magnetism experiments and magnetic mineralogy show that this characteristic magnetization is carried by Ti-poor magnetite or by hematite that replaces magnetite by late-magmatic deuteric alteration. Both magnetite and hematite carry the same characteristic component. The mean direction (Dm = 183.0°, Im = 53.5°, N = 10, α_{95} = 9.8°, K = 25.2) yielded a paleomagnetic pole located at 298.8°E, 63.3°S (α_{95} = 10.2°, K = 23.6), which is classified with a quality factor Q = 5. Paleogeographic reconstructions using this pole and other reliable Paleoproterozoic poles suggest that Laurentia, Baltica, North China Craton and Amazonian Craton were located in laterally contiguous positions forming a large continental mass at 1790 Ma ago. This is reinforced by geological evidence which support the existence of the supercontinent Columbia in Paleoproterozoic times.

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1. Introduction

Defining supercontinent paleogeographies is of outmost importance for understanding the mechanisms operating in the continental cycling of the early Earth (Condie, 2002). In recent years several studies have discussed the amalgamation and dispersion of large Paleo- to Neoproterozoic supercontinents (e.g., Rogers, 1996; Dalziel, 1997; Meert, 2002; Meert and Torsvik, 2003; Rogers and Santosh, 2003; Cordani et al., 2003; Pesonen et al., 2003; Zhao et al., 2005, 2006; Teixeira et al., 2007; Kusky et al., 2007). A Paleoproterozoic supercontinent has been envisaged by some authors but with significantly different paleogeographic configurations (e.g., Rogers, 1996; Meert, 2002; Rogers and Santosh, 2002; Zhao et al., 2002, 2003, 2004; Pesonen et al., 2003; Kusky et al., 2007). One of such paleogeographies joins the Amazonian Craton, Laurentia and Baltica through their Paleo to Mesoproterozoic belts, forming an elongated, continuous landmass named 'Columbia' (e.g., Rogers and Santosh, 2002; Geraldes et al., 2001; Pesonen et al., 2003). Paleomagnetism of well-dated geological formations from these cratonic units could be used to test the 'Columbia' hypothesis, however, Paleoproterozoic key poles for these cratons are still scarce (e.g., Meert, 2002; Pesonen et al., 2003). In this contribution we present a paleomagnetic data obtained on 1789 ± 7 Ma (U–Pb, zircon) felsic volcanics of the Colider Suite from northern Mato Grosso State, Amazonian Craton, Brazil. This well-dated Paleoproterozoic pole brings important constraints to the paleogeography of the Columbia supercontinent for which a new, paleomagnetic constrained configuration is proposed including Amazonia, Baltica, Laurentia and North China.

2. Geologic setting

The northeastern part of the Amazonian Craton is characterized by an Archean nucleus bordered by accretionary belts that are successively younger to the SW (Fig. 1), including: the 1950–1800 Ma Ventuari-Tapajós Province, the 1800–1550 Ma Rio Negro-Juruena Province, the 1500–1300 Ma Rondonia-San Ignacio Province, and the ca. 1250–900 Ma Sunsás Province (Tassinari et al., 2000; Cordani and Teixeira, 2007). The study area is situated in the southernmost part of the 1950–1800 Ma Ventuari-Tapajós Province (Tassinari

* Corresponding author. Fax: +55 11 30915034.
E-mail address: dagrella@iag.usp.br (M.S. D'Agrella-Filho).

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doi:10.1016/j.precamres.2008.03.004

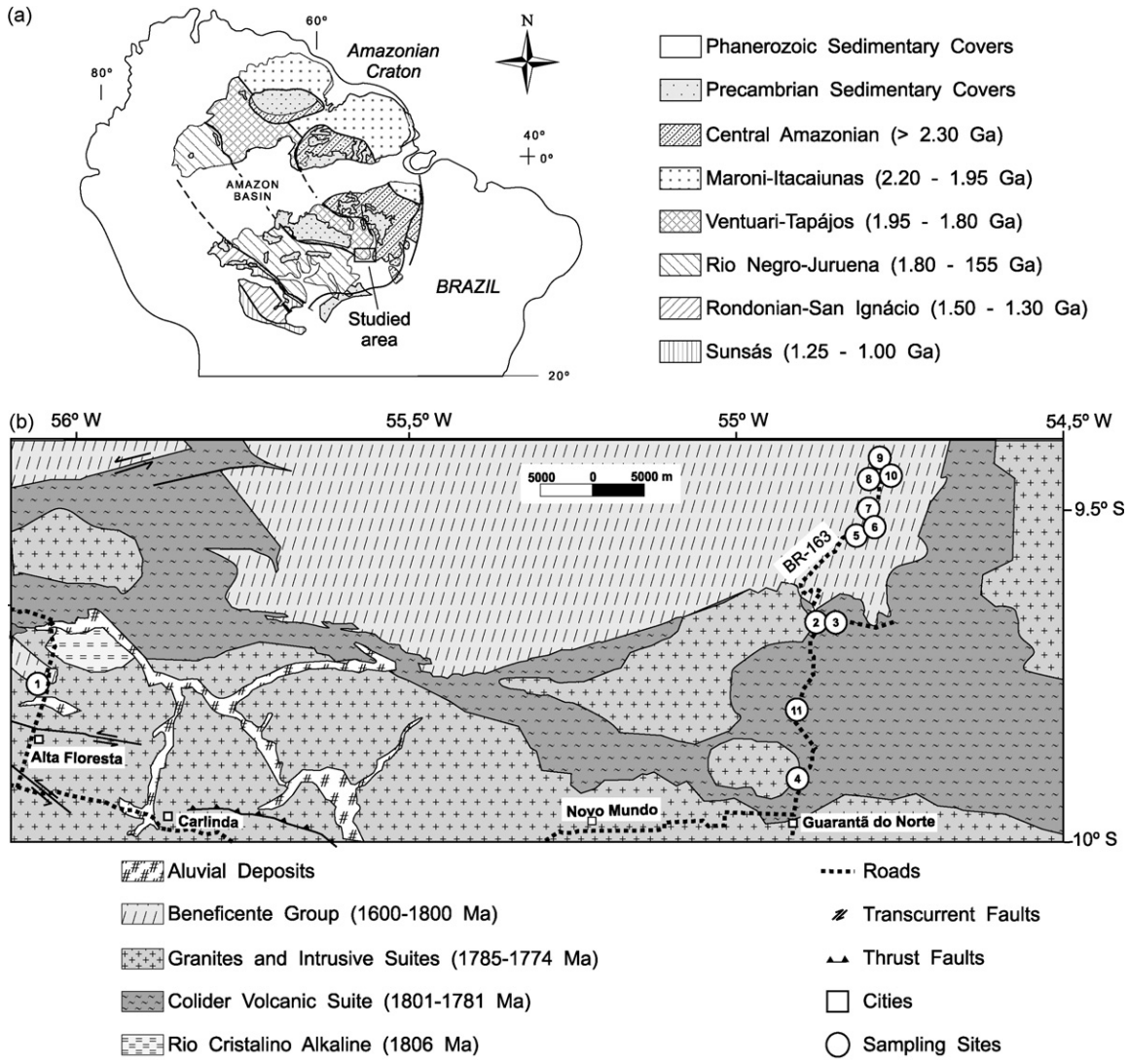


Fig. 1. (a) Amazonian Craton and their geochronological provinces (after Tassinari et al., 2000). (b) Geological map of the studied area with location of sampling sites (circled numbers) in the Colider region (after Lacerda-Filho et al., 2004).

and Macambira, 1999). This Province comprises granitic-gneissic rocks metamorphosed to amphibolite facies, with Rb–Sr and U–Pb zircon ages between 1950 and 1850 Ma. Sm–Nd model ages coincident or slightly older than their U–Pb and Rb–Sr radiometric ages, as well as positive $\varepsilon_{Nd(T)}$ values have been used to postulate a juvenile nature for these rocks (Cordani and Teixeira, 2007). This province is limited to the southwest by the Rio Negro-Juruena Province which is characterized by a younger 1800–1550 Ma juvenile mobile belt, comprising gneisses, granodiorites, tonalites, migmatites, granites, and amphibolites (Fig. 1). In the southwestern part of the area we find the Greenstone belt Alto Jauru composed of metavolcano-sedimentary sequences separated by granite-gneissic terranes of tholeiitic composition, intruded by dolerites, and partially covered by clastic rocks of the Aguapeí Group (~1200–1150 Ma).

The Iriri and Teles Pires Groups, comprising large felsic volcanic outcrops, were classically defined north of the Serra do Cachimbo region. U–Pb radiometric ages between 1870 ± 8 and 1890 ± 2 Ma were obtained for the Iriri Group by Vasquez et al. (1999), Lamarão et al. (1999), and Santos et al. (2000). The area of this study is located to the south of the Serra do Cachimbo region. Although initially correlated to the Iriri Group, the felsic volcanics cropping out in this region were shown to be about 100 Ma younger (Lacerda-Filho et al., 2004). Because of that, they were renamed as Colider Igneous Suite after the nearby city of Colider. The Colider Suite (Fig. 1) corresponds to high-K calc-alkaline acid to intermediate volcanic, subvulcanic, pyroclastic and epiclastic rocks, including rhyolites, riodacites and ignimbrites (Oliveira and Albuquerque, 2004). A mean age of 1789 ± 7 Ma is attributed to them, on the basis of two U–Pb TIMS (zircon) radiometric ages of 1781 ± 8 Ma (Pimentel, 2001) and 1801 ± 11 Ma (Pinho et al., 2001, 2003) and a U–Pb SHRIMP age of 1786 ± 17 Ma (Lacerda-Filho et al., 2004). Colider volcanics cover granite complexes, such as the 1872 ± 12 metaluminous Matupá Granites (Moura, 1998) and are covered by the siliciclastic sediments of the Paleoproterozoic Beneficente Group.

3. Sampling and methods

A total of 37 oriented block samples were collected from 10 sites comprising rhyolites, ignimbrites, and welded tuffs (Table 1 and Fig. 1). Samples were oriented using both sun and magnetic compasses. At the laboratory 3–4 cylindrical specimens 2.5 cm in diameter and 2.2 cm in height were cut from each block sample. Standard alternating field (AF) and thermal demagnetization

Table 1
Mean directions and poles obtained for Colider Suite acid rocks

Site	Localization	Samples	Site mean direction					VGP	
			N	Dec (°)	Inc (°)	α_{95} (°)	*K*	Plong (°E)	Plat (°N)
1	9.73°S/56.06°W	SD3-6	10	344.0	−60.8	3.3	215.3	144.9	55.1
2	9.63°S/54.88°W	SD15-17	11	176.6	58.9	2.8	264.1	310.3	−59.8
3	9.63°S/54.86°W	SD18-21	17	180.6	68.1	2.5	197.6	304.5	−48.4
4	9.91°S/54.91°W	SD26-33	7	350.5	−41.4	4.1	217.0	157.1	73.5
5	9.51°S/54.81°W	CS22-24	12	179.1	35.5	1.5	822.0	309.7	−79.8
6	9.51°S/54.79°W	CS25-27	12	205.8	45.8	3.6	142.7	254.7	−59.8
7	9.49°S/54.80°W	CS28-30	10	189.3	61.2	2.8	308.0	292.8	−56.2
8	9.42°S/54.81°W	CS31-33	7	198.9	63.7	2.6	540.0	284.2	−50.6
9	9.41°S/54.79°W	CS42-44	10	357.0	−27.3	5.1	90.2	154.6	84.1
10	9.42°S/54.78°W	CS45-47	11	196.9	65.3	2.8	265.7	287.6	−49.4
Mean			10	183.0	53.5	9.8	25.2		
CS pole						10.2	23.6	298.8	−63.3

N, Number of samples; Dec, declination; Inc, Inclination; α_{95}, *K*, Fisher's statistic parameters; Plong, pole longitude; Plat, pole latitude.

techniques were employed at the paleomagnetic laboratory of the University of São Paulo. A modified Magnetic Measurements MMTD60 furnace (peak temperature within ±2 °C, heating time of 1 h) was used for the stepwise thermal demagnetization. Remanent magnetizations were measured with a 2G-Cryogenic magnetometer or a Molspin spinner magnetometer. AF demagnetization was carried out by an automated three-axis AF-demagnetizer coupled with the cryogenic magnetometer. These instruments are housed in a magnetically shielded room with ambient field <1000 nT. Magnetic components for each specimen were identified in orthogonal plots, and calculated using the least squares fit method (Kirschvink, 1980). Fisher's (1953) statistics was used to calculate vector mean directions and the paleomagnetic pole.

For the magnetic mineralogy study, low-field magnetic susceptibility was measured during continuous heating and cooling of powder specimens up to 700 °C using a CS-3 apparatus coupled to the KLY-3 Kappabridge instrument (Agico, Czech Republic). Hysteresis curves were obtained with a Molspin Vibrating Sample Magnetometer (VSM), and isothermal remanent magnetization (IRM) curves for some specimens were performed using a pulse magnetizer (MMPM9, Magnetic Measurements). In addition, optical and scanning electron microscope observations (and EDS analyses) were performed on polished thin-sections from nine samples, corresponding to sites #1 to #4 (SD samples), which represent the rhyolites and ignimbrites of the Colider Suite.

4. Magnetic mineralogy

The microscopic characterization of opaque phases may be used to define their primary or secondary origin, thus it can help in the interpretation of the origin of their magnetic remanence. The main opaque phases in Colider Suite rocks comprise magnetite, Ti-poor magnetite and Ti-poor hematite (martite). Magnetite and Ti-poor magnetite occur as euhedral, octaedral grains whose sizes are mostly between 2 and 50 μm, sometimes showing embayment textures (Fig. 2A). They are usually devoid of inclusions and were probably formed at initial magmatic stages, but opaque grains composed uniquely of magnetite were observed only in sites #1 and #4. In most samples, magnetite grains are partially or totally replaced by hematite (Fig. 2B–D). In some polished sections, hematite is observed as tiny light-grey rims on brownish grey-colored magnetite (Fig. 2B and C). Hematite replacement is typically depleted in Ti relative to the magnetite precursor (Fig. 2D). In sites #2 and #3, opaque grains consist of Ti-poor hematite with octahedral shape with grain size (5–50 μm) in the same range of magnetite crystals (Fig. 2E). These hematite grains are interpreted as magnetite pseudomorphs (martite) formed by the complete replacement of former Ti-poor magnetite grains. Some hematite grains locally show secondary overgrowth, coating and filling-into pores around the opaque grain (Fig. 2F).

Samples from sites #1 and #4, for which most opaque grains are pristine magnetite and Ti-poor magnetite, show rapid decrease in intensity of remanence during AF demagnetization indicating the dominance of low coercivities typical of magnetite (samples SD3-C1 and SD32-B1 in Fig. 3a). Hysteresis cycles for the same samples show narrow waists (Fig. 4a) and the IRM curves show a sharp increase in magnetization at inducing fields below 200 mT (sample SD5-A1; Fig. 5). These characteristics are also typical of magnetite. But a small increase in remanence after 200 mT and up to 2 T suggests a low fraction of a high-coercive phase, possibly hematite as identified in rims of opaque grains in the petrographic studies (Fig. 2B and C).

Samples from sites #2, #3, as well as sites #5 to #10, for which hematite is the main opaque phase, are dominated by a high coercivity fraction which cannot be demagnetized even in fields as high as 160 mT (samples SD15-A and SD21A1 in Fig. 3a). Unblocking temperatures close to 660 °C in thermal demagnetizations from these sites suggest that the main remanence carrier is hematite (Fig. 3b). But small quantities of magnetite are also indicated by unblocking temperatures close to 580 °C (sample SD17-B2; Fig. 3b). Hysteresis curves and IRM acquisition curves support the interpretation of the AF and thermal demagnetization results. The wasp-waisted curves (Fig. 4b and c) are typical for mixtures between magnetic fractions with contrasting coercivities and they are interpreted as a result of the simultaneous presence of magnetite and hematite. Indeed, IRM curves for these samples are far from saturation in magnetic fields up to 2 T, suggesting that hematite is the dominant ferromagnetic mineral in these rocks (samples SD15-B1 and SD18-A1 in Fig. 5), but a small increase in remanence at low fields (150–200 mT) suggests also the presence of another, low-coercitive magnetic fraction, likely magnetite.

Thermomagnetic curves show irreversible behavior for all analyzed samples, being characterized by an increase in magnetic susceptibilities during cooling (Fig. 6). This suggests that during heating magnetite is formed, probably by the alteration of other non-magnetic minerals like micas and other Fe-bearing silicates. In samples from sites #1 and #4, heating and cooling curves present a steep decrease in susceptibility at temperatures around 580 °C (Fig. 6a), indicating the presence of magnetite. Hematite, with Néel/Curie temperature around 680 °C, was not observed in these curves probably due to the comparatively lower magnetic susceptibility associated to this mineral. The rocks at the other sites show strongly irreversible curves, indicating the formation of new mag-

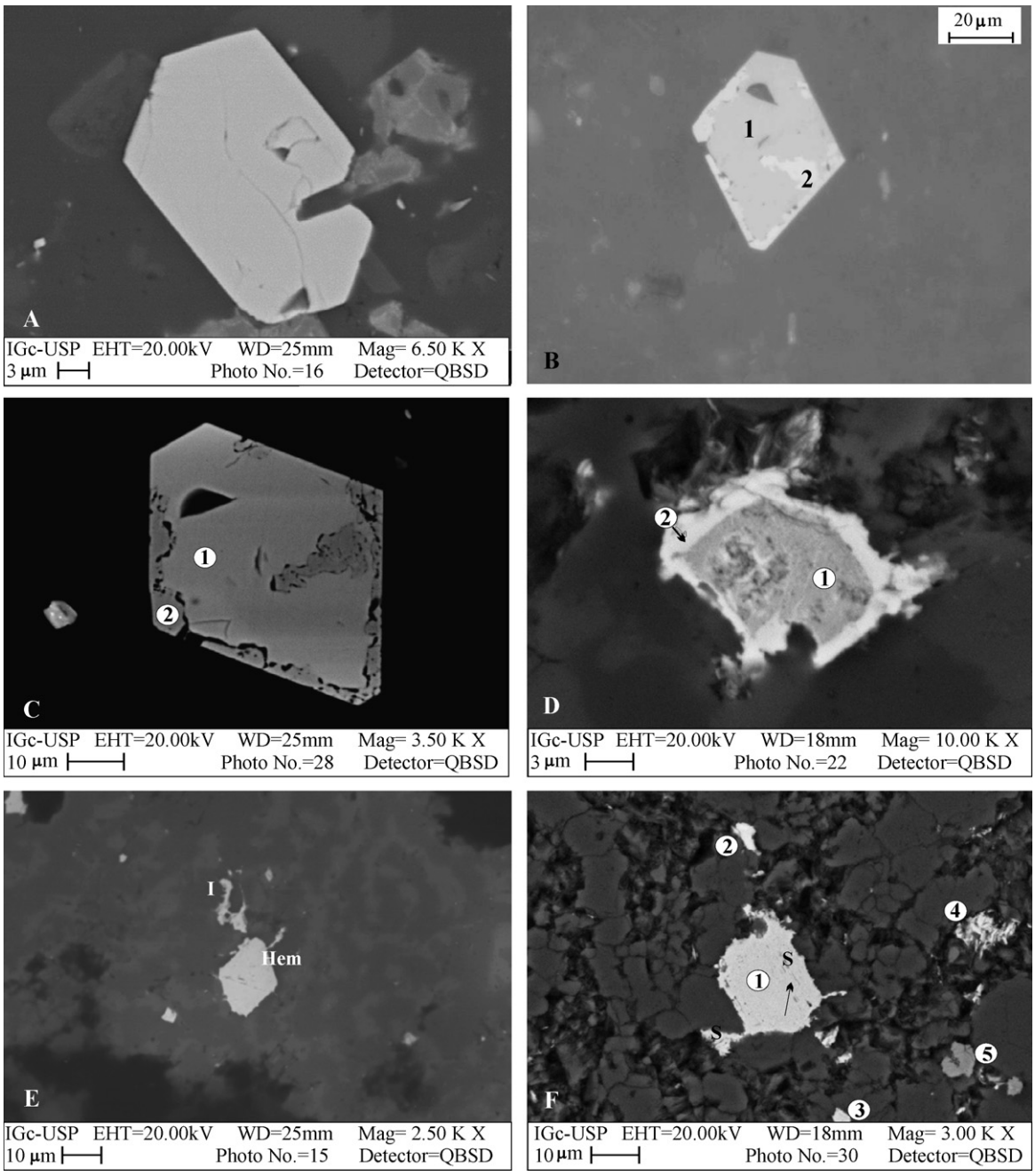


Fig. 2. (A) Electron SEM image of magnetite crystal with embayment texture (sample SD-5A2, site 1); (B) reflected light photomicrograph and (C) electron SEM image of a magnetite crystal with hematite rims (sample SD-4A1, site 1); (D) electron SEM image of a high-Ti magnetite (1), rimmed by hematite (2) (sample SD-18, site 3); (E) electron SEM image of hematite grain replacing magnetite (pseudomorph), with secondary overgrowth, coating and filling-into pores (sample SD-16A1, site 2); (F) electron SEM image of hematite grain (1) with secondary hematite overgrowth (S) (see arrow), zircon (2), titanite (5), and hematite (3 and 4) crystals. Contrast effect was enlarged in subparts (B–D).

netic phases at around 400 °C (Fig. 6b). Magnetic susceptibility in the cooling curve is significantly higher than in the initial steps of the heating curve.

5. Paleomagnetic results

Samples from sites #1 and #4 were easily AF demagnetized (Fig. 3a). Conversely, for the other sites thermal demagnetization was more efficient than AF demagnetization to isolate the characteristic remanent magnetization (ChRM) due to the significant amount of hematite in these rocks (Fig. 3a). For all sites a common characteristic component (ChRM) was isolated after elimination of a weaker, unstable directional component at fields lower than 30 mT and temperatures below 300 °C (Fig. 7). The characteristic magnetic component comprises a two-polarity north (south) direction with upward (downward) inclination.

Site mean ChRM directions are shown in Table 1 and Fig. 8. Normal and reverse directions pass a reversal test for common site-mean (McFadden and Lowes, 1981). The dual-polarity nature of the characteristic component and its moderate dispersion suggest that secular variation may have been averaged out despite the limited number of sites. After inverting the normal polarity direc-

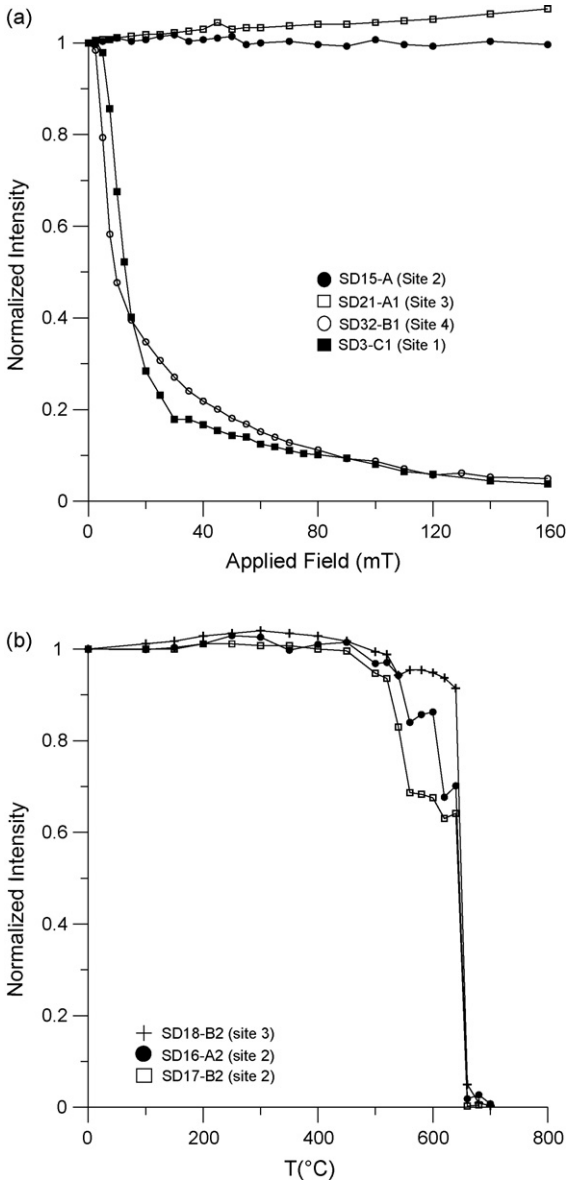


Fig. 3. Normalized intensities vs. (a) alternating magnetic field and (b) temperature for samples from different sites.

tions, a mean direction was calculated at Dm = 183.0°, Im = 53.5° (α_{95} = 9.8°, K = 25.2, N = 10) which yielded a paleomagnetic pole at 298.8°E, 63.3°S (α_{95} = 10.2°, K = 23.6). This pole, hereafter referred to as CS pole (Colider Suite pole), represents a well-dated 1790 Ma pole obtained on unmetamorphosed acid rocks. Using Van der Voo's (1990) criteria for paleomagnetic pole classification the Colider pole can be graded as Q = 5, since it has a (1) good age constraint, (2) satisfactory statistical parameters for the mean magnetization, (3) the characteristic magnetization was isolated after detailed demagnetization and principal component analysis, (4) there is adequate tectonic control of the sampled area, and (5) the magnetic record comprises reversals.

Petrographic observations suggest a primary origin for the characteristic component. In thin-section, the felsic volcanic rocks of Colider show pristine igneous textures without evidence of metamorphism or weathering. Magnetite crystals keep their octahedral forms, but they are usually rimed or replaced by hematite. Both primary magnetite and hematite replacements carry two opposite normal/reversed directions with unblocking tempera-

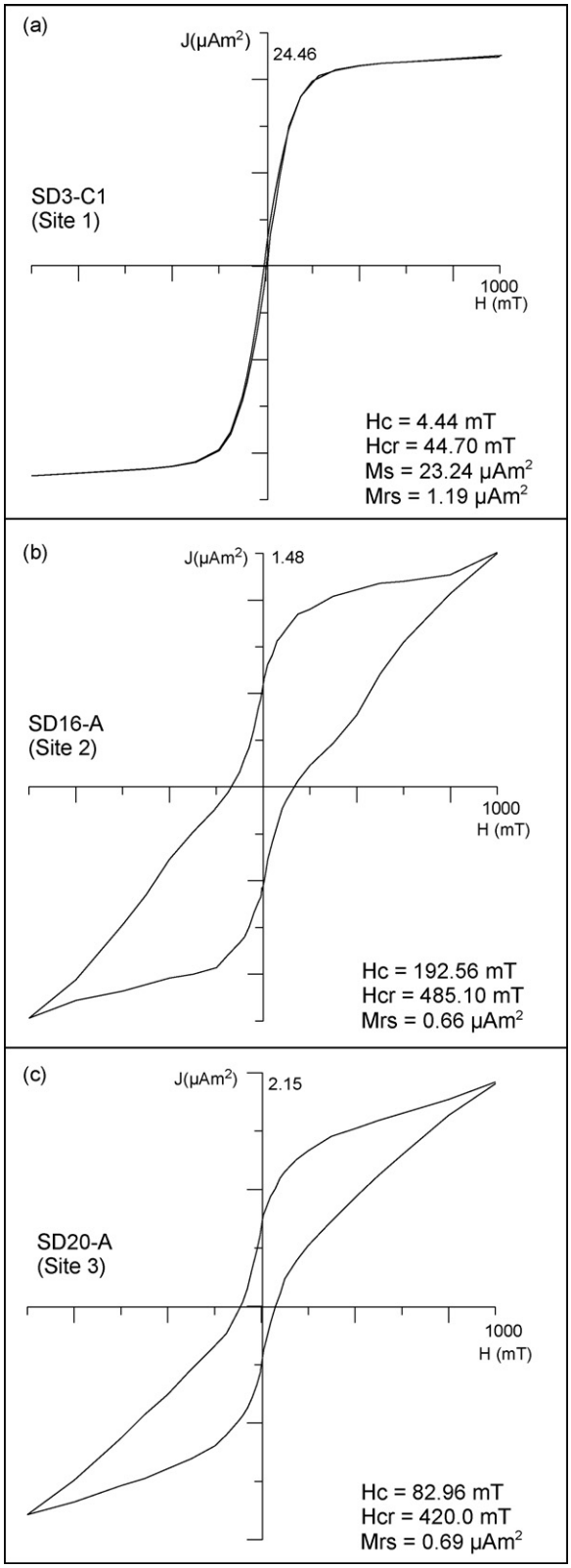


Fig. 4. Hysteresis curves indicating different degrees of mixture between magnetite and hematite. (a) narrow waist curve; (b) and (c) wasp-waisted curves.

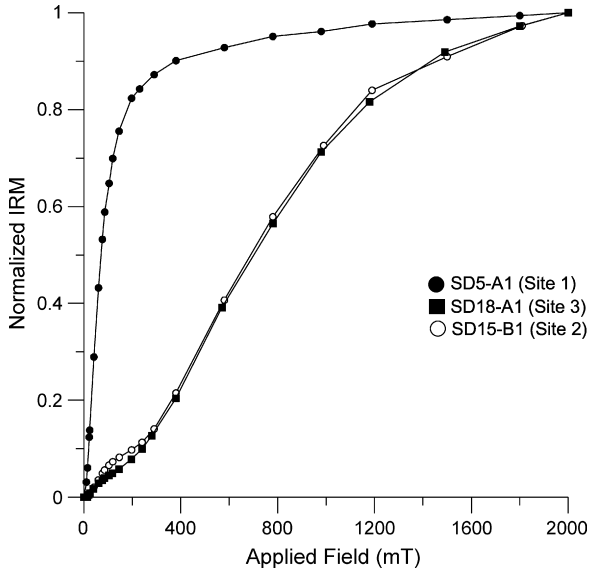


Fig. 5. IRM acquisition curves (normalized intensities vs. magnetic field).

tures very close to the Curie temperature of these minerals. Similar parageneses are described elsewhere in felsic volcaniclastic complexes, the hematite formation being interpreted as a result of late-magmatic oxidation of magnetite at high-temperature (e.g., Seaman et al., 1991; Alva-Valdivia et al., 2003; Saito et al., 2003).

6. Discussion

6.1. Building Columbia

Paleoproterozoic orogenic belts are found in several cratonic blocks with ages clustering at around 2100–1800 Ma, thus leading some authors to suggest the existence of a Paleoproterozoic supercontinet (e.g., Rogers, 1996). Most of the Paleoproterozoic reconstructions were based on geological/geochronological similarities between cratons, hence given the high degree of freedom in connecting cratonic pieces together a myriad of paleogeographies have been proposed (e.g., Rogers, 1996; Geraldes et al., 2001; Rogers and Santosh, 2002; Kusky et al., 2007). Among these reconstructions, one comprising a long continuous landmass formed at 1800 Ma by Laurentia, Baltica and Amazonia connected through their Paleo- to Mesoproterozoic orogenic belts is of particular interest here. This configuration has been proposed on geological grounds (Geraldes et al., 2001) and further tested by Pesonen et al. (2003) using key poles from Laurentia and Baltica; no key-poles were available for Amazonia by that time. The Colider Suite pole presented here may be used to further test such a paleogeographic configuration.

In Fig. 9 we present our paleographic reconstruction for 1790 Ma. It follows the configuration previously proposed (Geraldes et al., 2001 and Pesonen et al., 2003) but also includes the North China Craton that fills the 20–30° gap between Amazonia and Baltica. The addition of the North China Craton in Columbia reconstructions was recently proposed by Kusky et al. (2007). Our reconstruction is based on reliable paleomagnetic poles presented in Table 2. It places North China Craton in a slightly different position (rotated by ca. 90° clockwise) from that of those authors. A brief discussion on the choice of paleomagnetic poles and the geological evidence of the suggested position of each craton in Columbia is presented below.

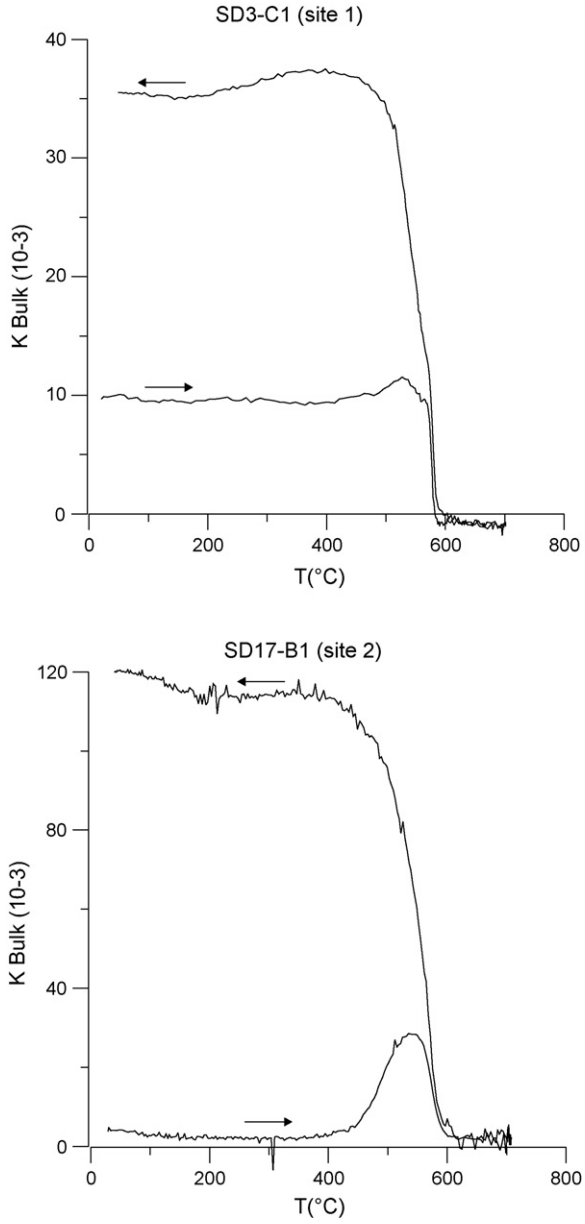


Fig. 6. Typical thermomagnetic curves showing variation in magnetic susceptibility (SI) with temperature. Curves were corrected from furnace effects.

6.2. Amazonian Craton in Columbia

The position of the Amazonian Craton is constrained by the well-dated 1790 Ma Colider Suite pole (CS pole) obtained on felsic volcanic rocks covering the Ventuari-Tapajós province (Fig. 1). The Ventuari-Tapajós Province is characterized by a voluminous juvenile granitic magmatism of calc-alkaline composition between 1980 and 1810 Ma (Cordani and Teixeira, 2007). In the southern part of the province, these plutons intrude low-grade supracrustal sequences comprising detrital zircons with ages varying from 3100 Ma up to 1900 Ma (Santos, 2003). These rocks are covered by the anorogenic volcanism of the Colider Suite and the continental siliciclastic sediments of the Beneficente Group, marking the cratonization of the Ventuari-Tapajós Province coeval to the development of the Rio Negro-Juruena magmatic arcs in the southwest (Cordani and Teixeira, 2007).

Table 2
Selected paleomagnetic poles for the 1790 Ma paleogeographic reconstruction of Fig. 9

Continent	Formation	Plat (°N)	Plong (°E)	α_{95} (°)	Age (Ma)	Reference
Laurentia	Dubawnt Group	7.0	277.0	8.0	1785 ± 4	Park et al. (1973)
Baltica	Shosksha Formation	42.0	221.2	7.0	1790–1770	Pisarevsky and Sokolov (2001)
Baltica	Ropruchey Sills	40.5	229.8	8.1	1770 ± 12	Fedotova et al. (1999)
North China	North China Dikes	36.0	247.0	3.0	1769.1 ± 2.5	Halls et al. (2000)
Amazonian Craton	Colider Suite	−63.3	298.8	10.2	1789 ± 7	This work

Plat, Pole latitude; Plong, pole longitude; α_{95}, Fisher's statistic parameter.

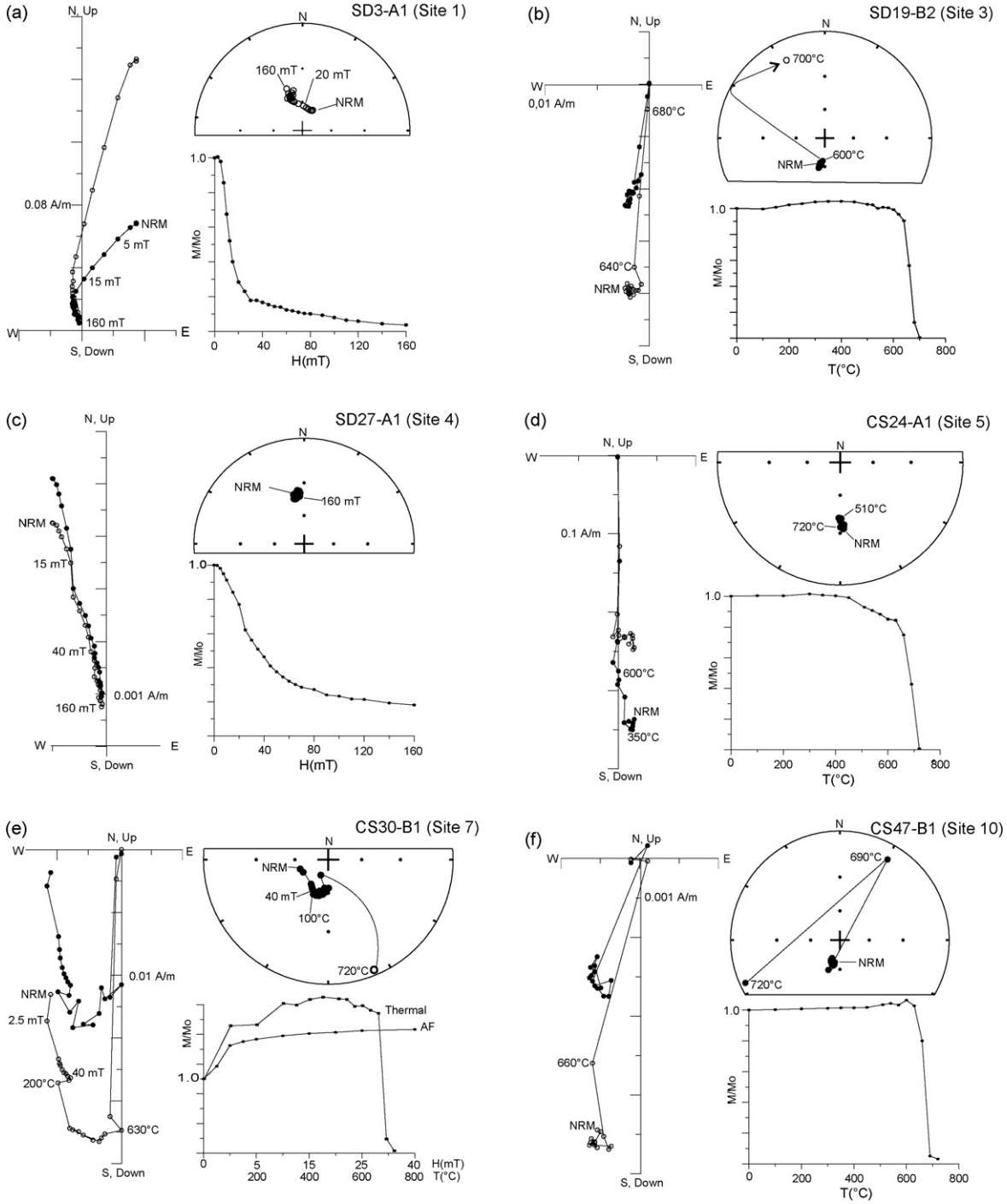


Fig. 7. Examples of AF (a, c), thermal (b, d, f) and mixed (e) demagnetization. Orthogonal projections: full (empty) circles represent horizontal (vertical) projections. Stereographic projections: full (empty) circles represent downward (upward) inclinations. Normalized intensity curves.

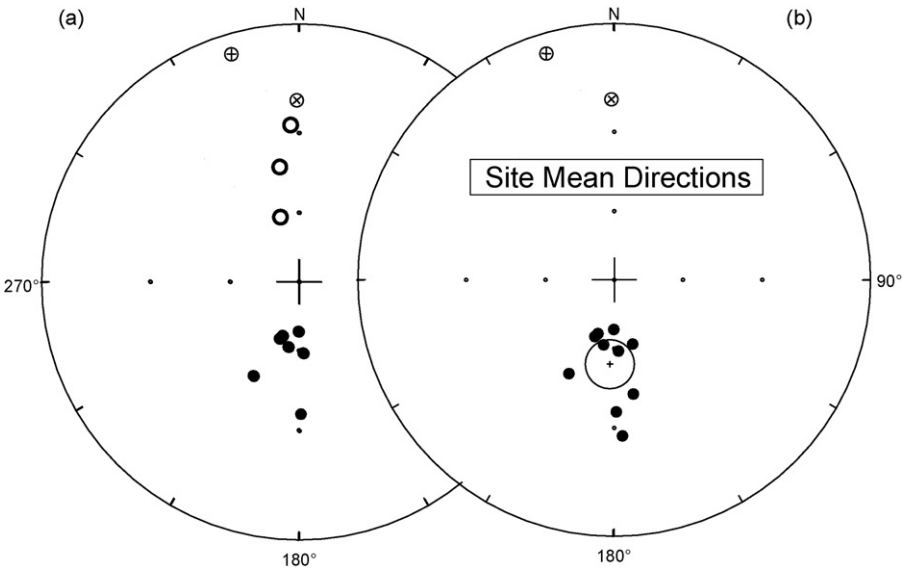


Fig. 8. Site mean directions (a) normal and reverse directions and (b) after inversion of normal polarity directions. Full (empty) circles represent downward (upward) inclinations. Symbols (⊗ and ⊕) represent the present dipolar and the actual geomagnetic field, respectively.

6.3. Baltica in Columbia

The paleogeography of Baltica is based on the mean of two key poles: the 1770 Ma Ropruchey Sills and the ca. 1790 Ma Shosksha Formation. The first one is obtained from well-dated 1770 ± 12 Ma (U–Pb) gabbros and dolerites from the Ropruchey Sills whose magnetizatizion is considered primary in origin (Fedotova et al., 1999). The second one is determined from sandstones from the top of the Shosksha Formation (Pisarevsky and Sokolov, 2001). These sediments are younger than granites with ages of 1794 ± 24 and 1778 ± 16 Ma (U–Pb in zircon), and they are intruded by the 1770 Ma Ropruchey Sill. So, we can attribute an age between 1790 and 1770 Ma for these sediments. The stable remanence isolated in these rocks is carried by single domain early diagenetic hematite (Pisarevsky and Sokolov, 2001). These sedimentary and shallow intrusive rocks post-date the amalgamation of Karelia and Kola Cratons in Baltica, that occurred at 1950–1820 Ma (e.g., Gorbatschev and Bogdanova, 1993), forming the Lapland-Kola Belt (Fig. 9). The Paleoproterozoic Svecofennian domain situated west and southwest of the Karelia Craton is regarded as a juvenile Paleoproterozoic accretionary belt, evolving at about the same time as the Kola-Karelia collision (Buchan et al., 2000).



Fig. 9. Paleogeographic reconstruction for 1790 Ma using poles from Table 2. Archean cratonic areas (light-grey): Laurentia (S, Slave; C, Churchill; SU, Superior; N, Nain and equivalent in southern Greenland), Baltica (KO, Kola; KA, Karelia), North China (YB, Yinshan Block; OB, Ordos Block; WB, West Block; EB, East Block), Amazonia (CA, Central Amazonia Province). Paleoproterozoic belts (dark grey): Laurentia (NQ, New Quebec; T, Torngat; W, Wopmay; P, Penokean; K – Kefilidian; NA, Nagssugtoqidian; FR, Foxe-Rinklan), Baltica (LK, Lapland-Kola; SD, Svecofennian Domain), North China (NH, North Hebei/Khondolite belt; TNC, Trans-North China), Amazonia (MI, Maroni-Itacaiunas; VT, Ventuari-Tapajós; RNJ, Rio Negro-Juruena).

6.4. Laurentia in Columbia

For Laurentia, the available paleomagnetic poles with ages between 1770 and 1790 Ma permit ambiguous interpretations (see Pesonen et al., 2003). Poles obtained for the Trans-Hudson Orogenic Province (THO) suggest high paleolatitudes for Laurentia, whereas poles from cratonic provinces (Superior, Churchill and Coronation) indicate moderate paleolatitudes. The 1830 Ma Molson dikes (Superior Craton) yield three components but the primary or secondary nature of these components is debatable (Zhai et al., 1994; Halls and Heaman, 2000). The Peninsular sills (Coronation Province) yield well-clustered low-inclination remanence directions (Irving and McGlynn, 1979), and the calculated pole has been used by Pesonen et al. (2003) for a 1770 Ma reconstruction. However, this pole cannot be considered a key pole due to its inconclusive baked contact test and the poor age constraints (a Rb–Sr isochron age of 1800 Ma). Park et al. (1973) obtained paleomagnetic data for sedimentary rocks of the Kazan Formation, volcanic rocks from Christopher Island Formation (Dubawnt Group-Churchill province) besides two large dikes considered as feeding dikes of the Christopher Island lava flows. All these rocks presented consistent results that passed reversal and baked contact tests. K–Ar dating in biotite and phlogopite of the igneous rocks yielded a mean age of 1716 Ma, but more recent studies attribute an older age, between 1830 and 1760 Ma, for the Dubawnt Group (Rainbird and Hadlari, 2000). Based on these results, we used the Dubawnt pole to constrain the Laurentia position at ~1780 Ma (Fig. 9), although we are aware that other interpretations are also possible. By that time,

Laurentia has already been amalgamated by the collision of Superior, Slave, Nain, and Churchill Archean cratonic blocks (Hoffman, 1989).

6.5. North China in Columbia

The inclusion of the North China Craton in Paleoproterozoic reconstructions is the result of the improved knowledge obtained on the Precambrian geological evolution of this region (e.g., Zhai and Liu, 2003; Zhao et al., 2006; Kusky and Li, 2003; Kusky et al., 2007; Liu et al., 2006). In here, the North China Craton was positioned using the key paleomagnetic pole obtained for the 1769 ± 2.5 Ma (U–Pb, zircon) Fuping dike swarms by Halls et al. (2000). The Paleoproterozoic evolution of North China bears large similarity to that of Baltica and the Amazonian Craton. A possible connection between North China Craton and Baltica has been proposed by Qian (1997), based on lithological and geochronological correlations. Wilde et al. (2002) also suggested the conecction between North China and Baltica based on correlations of Archean cratonic areas and Paleoproterozoic collisional zones. Recently, Kusky et al. (2007) proposed a new configuration for the Columbia Supercontinent where the North China Craton was connected to Baltica and Amazonian Craton between 2000 and 1800 Ma. In his model, based on the fit of orogenic belts, Baltica Svecofennian domain is connected to North China through the Paleoproterozoic North Hebei belt. In our paleomagnetically constrained reconstruction, the North China links the Paleoproterozoic belts of Baltica and Amazonia through the Trans-North China belt instead (Fig. 9). This paleogeography agrees with Zhao et al. (2005, 2006) model of the North China amalgamation where the Trans-North China belt resulted from the collision between the Eastern and the Western blocks (Fig. 9) at Paleoproterozoic times (1850 Ma).

6.6. Columbia disruption

During the Paleoproterozoic, an eastward Andean-type subduction-related process was in progress along the southwestern border of the Amazonian Craton and the Eastern Block of North China (Fig. 9). In Amazonia, the juvenile magmatic arcs (Ventuari-Tapajós) were being formed between 1950 and 1800 Ma (Sadowski and Bettencourt, 1996; Geraldes et al., 2001; Tassinari and Macambira, 2004; Cordani and Teixeira, 2007), whereas in North China the Western and Eastern Blocks collided at 1850 Ma. These processes led to the development of the Ventuary-Tapajós and the Trans-North China belts. As a result of these accretions and other collisional processes, by ∼1800 Ma ago the Columbia supercontinent amalgamated Laurentia, Baltica, North China Craton, Amazonian Craton, and probably many other cratons (Rogers, 1996; Meert, 2002; Meert and Torsvik, 2003; Rogers and Santosh, 2002, 2003). However, soon after Columbia assembly, several extensional processes have developed in the North China and Amazonian Cratons. In North China the tectonic regime at 1800–1770 Ma is characterized by the development of several aulacogen systems, associated with a voluminous felsic extrusive magmatism, and with profuse emplacement of predominantly NW to NE trending mafic dike swarms, as well as anorogenic intrusions which include rapakivi granites, anorthosites, gabbros, ultramaphic rocks and isolated granites and pegmatites (Kusky et al., 2007). Interestingly, at the same time the Amazonian Craton was intruded by a voluminous magmatism comprising mafic dike swarms in Venezuela, Roraima (Avanavero dikes) and Mato Grosso States from Brazil dated at 1780 Ma (Santos et al., 2003) followed by Mesoproterozoic anorogenic rapakivi granites (Sadowski and Bettencourt, 1996; Santos et al., 2000; Geraldes et al., 2001). Kusky et al. (2007) associate this tectonic regime to an initial rupture of the Columbia Supercontinent, with the origin of passive margins at the present southwestern and northern margins of the North China Craton at that time. Following this interpretation, Columbia would be an ephemeral supercontinent.

7. Conclusions

A paleomagnetic study of the well-dated 1789 ± 7 Ma unmetamorphosed felsic volcanic rocks from the Colider Suite (Amazonian Craton) yields a paleomagnetic pole (SC pole) located at 298.8°E, $I = 63.3°$S ($\alpha_{95} = 10.2°$, $K = 23.6$). Magnetic mineralogy and petrographic analyses suggest that the SC pole represents a primary thermochemical remanent magnetization, and it is classified with a reliability factor $Q = 5$. A paleogeographic reconstruction for 1800–1780 Ma ago (Fig. 9), obtained using reliable paleomagnetic poles for Laurentia, Baltica, North China Craton and Amazonian Craton (CS pole), permit to display these cratons in a laterally contiguous configuration which characterizes the final amalgamation of the Paleoproterozoic Columbia supercontinent. The geological evidence from North China craton might suggest that this supercontinent was ephemeral, being disrupted soon after its formation.

Acknowledgements

We thank Maria Irene Bartolomeu Raposo (IGc-USP, Brazil) for the permission to do hysteresis measurements in the IGc Anisotropy of Magnetic Susceptibility Laboratory. Companhia Matogrossense de Mineração (METAMAT) helped in field logistics. SEM analyses were made in the SEM laboratory of the IGc-USP (95/5635-4). This work received financial support from FAPESP (grant 03/12802-2) and CNPq (grant 55.4458/05-5). We thank Joe Meert and Peter Cawood for their insightfull reviews of this paper.

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